LEARNER GUIDE

# Where two lines meet

Equations, Inequalities and Simultaneous Relationships · Lesson 6. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

## What this lesson is about

A single line is a whole set of solutions. Take **two** lines and ask: which pair $(x, y)$ satisfies *both* equations at once? Since each line is its own equation's solution set, a pair on **both** lines must be where they **cross**. That crossing is the **common solution** — the answer to the two equations taken together (solved **simultaneously**). You can read it off the graph and confirm it by substituting into each equation: the common solution is the one point that makes both true.

## What you are learning to notice

- The point where two lines cross lies on **both** lines.
- It is the **common solution**: it satisfies both equations at once.
- Points on just one line satisfy only that equation — not both.
- Check the crossing by substituting into *each* equation; both must come out true.

## Moving through it, one step at a time

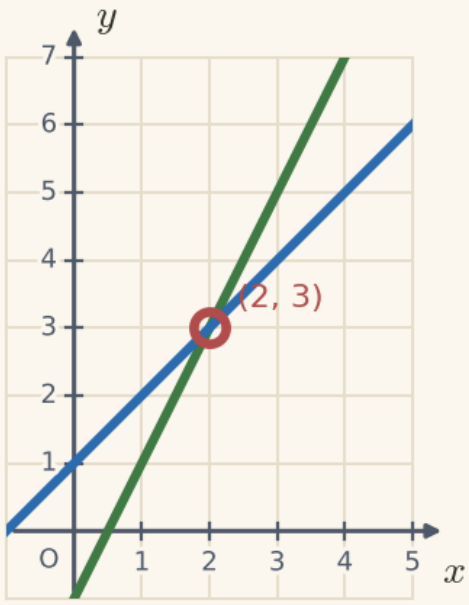
Plot both lines on the same axes and read the point where they cross. To be sure, substitute that point into each equation in turn — it must satisfy both. Algebraically, the crossing is found by setting the two right-hand sides equal (because at the crossing the $y$-values agree), solving for $x$, then finding $y$. Either way, one pair answers both equations together.

## Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

**Example 1 — Reading the common solution**

Where do $y = 2x - 1$ and $y = x + 1$ cross?

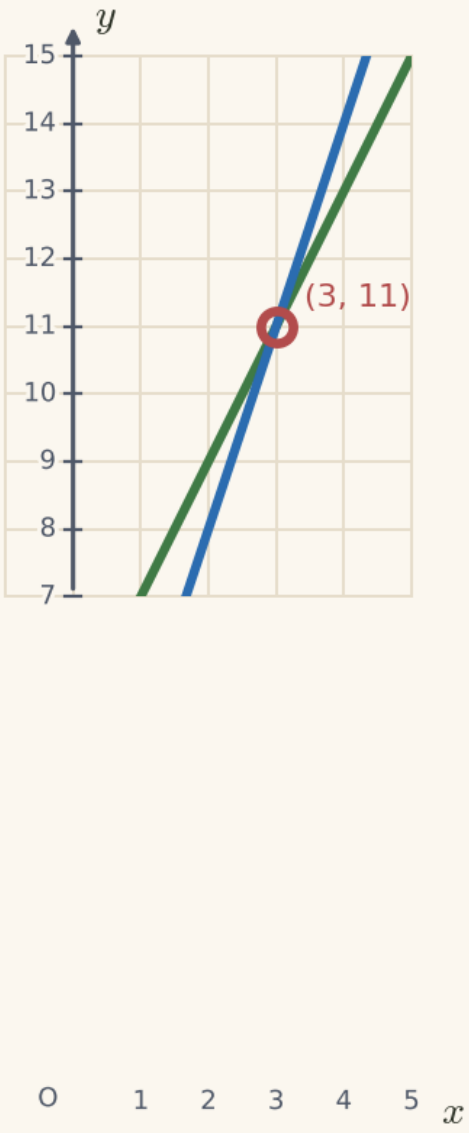



| | |
|---|---|
| **1.** At the crossing the $y$-values agree, so set the sides equal. | $2x - 1 = x + 1$ |
| **2.** Collect the $x$-terms and solve. | $x = 2$ |
| **3.** Find $y$ from either equation. | $y = 2 + 1 = 3$ |
| **4.** Check in both. | $2\times2 - 1 = 3$ ✓ and $2 + 1 = 3$ ✓ |

**(2, 3) — the crossing satisfies both equations, so it is the common solution.**

**Example 2 — A break-even point**

Plan A costs $y = 2x + 5$; Plan B costs $y = 3x + 2$. When do they cost the same?




| | |
|---|---|
| **1.** Set the two costs equal. | $2x + 5 = 3x + 2$ |
| **2.** Solve for $x$ (the hours). | $x = 3$ |
| **3.** Find the shared cost. | $y = 2\times3 + 5 = 11$ |
| **4.** Interpret the decision. | Same at (3, £11); cheaper plan differs either side |

**They break even at (3, £11) — the common solution. Before 3 hours one plan is cheaper; after, the other.**

**A gentle reminder**

You can pause. You can return. You do not need to complete everything in one sitting.

## If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading a line as a set of solutions (Lesson 5), reading coordinates, and substituting a point), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

**How the worked solution unlocks**

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

## Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

## Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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